1 Arithmetic and Geometric Progressions

An arithmetic progression is defined by its starting term 'a_1' and unique common difference 'd'. A general AP can be written as a_1, a_2,..., a_n. Some useful results follow.

- Notice that the general AP can also be written as a_1, $a_1 + d$, $a_1 + 2d$,..., $a_1 + (n-1)d$.

$$a_n = a_1 + (n-1)d \tag{1}$$

- To sum the general AP, we can notice that the sum of the first and last terms is equal to the second and second-last terms and so on which is equal to $2a_1 + (n-1)d$ and we have $\frac{n}{2}$ pairs in the general AP.

$$S_n = \frac{n}{2}[2a_1 + (n-1)d] = \frac{n}{2}[a_1 + a_n] \tag{2}$$

A geometric progression is defined by its starting term 'u_1' and unique common ratio 'r'. A general GP can be written as u_1, u_2,..., u_n. Some useful results follow.

- Just like above, the general GP can be wriiten in a different way.

$$u_n = u_1 \cdot r^{n-1} \tag{3}$$

- To sum the general GP, we can shift the GP to the right and make its starting term u_2 by multiplying the entire GP by r. If we substract both these GPs we get a telescoping series.

$$S_n = u\left(\frac{1-r^n}{1-r}\right) = \frac{u_1 - u_n r}{1-r} \tag{4}$$

- Notice that if $|r| < 1$, the infinite sum seems to sum to a finite number. We can argue that as the value of n increases, the value of r^n decreases i.e. the limit r^n approaches as n goes to infinity is zero. None of the other terms in (4) care what the value of n is.

$$S_\infty = \frac{u}{1-r} \tag{5}$$

1. We can say three terms form an AP if $a_3 - a_2 = a_2 - a_1$. It comes out to be equal to the following term.

$$\frac{2\sqrt{b} - \sqrt{a} - \sqrt{c}}{2d}$$

Hence the new terms form an AP.

2. Notice that the sum can be written in an another way which gives us d in the denominator. This is a telescoping series where after simplification we end up with the following.

$$\frac{n-1}{\sqrt{a_1} + \sqrt{a_n}}$$

3. Same trick here. If we multiply with d, we get a telescoping series which after simplification gives us the following.

$$\frac{n-1}{a_1 a_n}$$

4. Follows from above. This telescoping series trick is only possible with the proposition that $a_1, a_2, ..., a_n$ are in an AP.

5. Let P(n) denote the sum given.

$$P(n) = \sum_{i=0}^{n} (-1)^i C_n^i a_{i+1} = 0$$

We know that $a_{i+1} = a_i + id$. So the sum can be written as follows.

$$\sum_{i=0}^{n} (-1)^i C_n^i [a_1 + id] = \sum_{i=0}^{n} (-1)^i C_n^i a_1 + \sum_{i=0}^{n} (-1)^i C_n^i id$$

From summation properties and absorption/subset identity ($k \cdot C_n^k = n \cdot C_{n-1}^{k-1}$) , we can simplify as follows.

$$a_1 \cdot \sum_{i=0}^{n} (-1)^i C_n^i + nid \cdot \sum_{i=1}^{n} (-1)^i C_{n-1}^{i-1}$$

By change of index from i to $k = i - 1$ in the second sum we can write the total sum as follows.

$$a_1 \cdot \sum_{i=0}^{n} (-1)^i C_n^i + nid \cdot \sum_{i=0}^{n-1} (-1)^{i+1} C_{n-1}^i = a_1 \cdot \sum_{i=0}^{n} (-1)^i C_n^i - nid \cdot \sum_{i=0}^{n-1} (-1)^i C_{n-1}^i$$

By invoking the binomial theorem, we get the following result

$$a_1 \cdot 0 - nid \cdot 0 = 0 = P(n)$$

Hence, the proposition P(n) is true for all $n \geq 2$ as $LHS = RHS$. Hence for any AP we have the equalities given by P(n).

6. Same trick as above using the following property $k(k-1)C_n^k = n(n-1)C_{n-2}^{k-2}$.

7. Solve this using the given equality $\frac{1}{logk} + \frac{1}{logn} = \frac{2}{logm}$ as well as logarithmic properties. Take the log of both sides and transfer $logn$ to the right and then show LHS = RHS.

8. You have to show that the ratio $\frac{S_n}{S_{kn} - S_n}$ is independent of n and you will find it to be equal to $\frac{1}{k^2-1}$.

9. Starting with the first equation, using some manipulation of terms we can show,

$$x_1 = \frac{a}{n} - \left(\frac{n-1}{2}\right) d$$

Then in the second equation using $a_n = a_1 + kd$, we find the following (assuming $k = n - 1$),

$$\sum (x + kd)^2 = x^2 \sum 1 + d^2 \sum k^2 + 2xd \sum k$$

Simplifying this using $\sum a^2 = \frac{n(n+1)(2n+1)}{6}$ and $\sum a = \frac{n(n+1)}{2}$ and banishing k from above,

$$nx^2 + d^2 \left(\frac{n(n-1)(2n-1)}{6}\right) + xdn(n-1)$$

Substituting the value of x_1 we found earlier and simplifying gives us the following,

$$d = \pm \frac{2}{n} \sqrt{\frac{3(nb^2 - a^2)}{n^2 - 1}}$$

10. The number sequence corresponds to starting term 3 and common difference 3 so the nth term is $3n$ and sum of first n terms is $\frac{3n(n+1)}{2}$.

11. We know that the square of an odd natural number is given by $(2n+1)^2$ where n is any whole number. Assume an AP 1,2,3,...,n where $a = 1, d = 1$. Then each row is given by the following relation,

$$S_{3n+1} - S_n$$

Simplifying this with the sum of general AP formula we get

$$S_{3n+1} - S_n = (2n+1)^2$$

12. Notice if we divide a_{m+n} by a_{m-n} we get,

$$r^{2n} = \frac{A}{B} \implies r = \left(\frac{A}{B}\right)^{\frac{1}{2n}}$$

Also notice that if we multiply a_{m+n} and a_{m-n} we get,

$$a_m = \sqrt{AB}$$

Let $a = \frac{A}{r^{m+n-1}}$ then we can write a_n as

$$a_n = \frac{A}{r^m} = A\left(\frac{B}{A}\right)^{\frac{m}{2n}}$$

13. Notice that the sum $S_{2n} - S_n = a_{n+1} + a_{n+2} + ... + a_{2n} = r^n(a_1 + a_2 + ... + a_n) = r^n S_n$ and similarly $S_{3n} - S_{2n} = r^{2n} Sn$. Hence

$$\frac{S_n}{S_{2n} - S_n} = \frac{S_{2n} - S_n}{S_{3n} - S_{2n}} = \frac{1}{r^n}$$

14. Using the sum formula of a general GP we find that,

$$S_n = \frac{u(1-r^n)}{1-r}, \bar{S}_n = \frac{r^{1-n}}{a}\left(\frac{r^n - 1}{r-1}\right) \implies \frac{S_n}{\bar{S}_n} = a^2 r^{n-1}$$

Using $\sum_0^n i = \frac{n(n+1)}{2}$ and rules of exponents we can conclude that,

$$\prod = a^n r^{\frac{n(n-1)}{2}}$$

By observation we can conclude that

$$\left(\prod\right)^2 = \left(\frac{S_n}{\bar{S}_n}\right)^n$$

15. Let S be the required sum. Multiply S with x. We have a partial telescoping series which gives us

$$S - xS = (1 + x + x^2 + ... + x^n) - (n+1)x^{n+2}$$

After simplification and solving for S we get

$$S = \frac{1 - (n+2)x^{n+1} - (n+1)x^{n+2}}{(1-x)^2}$$

16. Let a_i represent a term in the sequence 1, 11, 111, ..., 111...1. Then we observe that: $\sum a_1 = 1$, $\sum a_2 = 12$, $\sum a_3 = 123$ and so on. Thus

$$\sum a_n = 123...n$$

17. Notice that we can write the series as:

$$nx + (n-1)x^2 + (n-2)x^3 + ... + 3x^{n-2} + 2x^{n-1} + 1x^n$$

$$(x^n + x^{n-1} + ... + x^2 + x) + (x^{n-1} + x^{n-2} + ... + x) + ... + (x^2 + x) + (x)$$

Notice that the sum of the given series is the sum of the same GP of terms n, n-1, ... 2, 1. Using the sum of GP formula we get:

$$\sum_1^n x\left(\frac{x^i - 1}{x-1}\right) = \frac{x}{x-1}\left(\sum_1^n x^i + \sum_1^n 1\right) = \frac{x^2(x^n - 1) - nx(x-1)}{(x-1)^2}$$

18. Let us denote the required sum as follows:

$$\mathbb{S} = \sum_{1}^{n} \frac{2i-1}{2^i} = \sum_{1}^{n} \frac{2i}{2^i} - \sum_{1}^{n} \frac{1}{2^i}$$

Using appropriate techniques we get the required sum:

$$\mathbb{S} = 3 - \frac{2n+3}{2^n}$$

19. Let a number in the sequence 49, 4489, 444889, ... be denoted by N_k. We claim that N_k is a perfect square for all k=0,1,2,... i.e. it is of the form $(number)^2$.

$$N_k = \underbrace{4...4}_{k}\underbrace{8...8}_{k-1}9 = \frac{4(10^k-1)}{9} \cdot 10^k + \frac{8(10^{k-1}-1)}{9} \cdot 10 + 9$$

If we take the common denominator over 9 and simplify we get the following for all k and N_k.

$$N_k = \left(\frac{2 \cdot 10^k + 1}{3}\right)^2$$

We can show that our base is a whole number. $10^k \equiv 1 \pmod 3 \implies 2 \cdot 10^k + 1 \equiv 0 \pmod 3$ meaning that $2 \cdot 10^k + 1$ is divisible by 3 and hence a whole number.

20. Let us denote any term in the geometric progression as a_i and the sum of subsequent terms as S.

$$a_i = q^{i-1} \quad \text{and} \quad S = S_\infty - S_i = \frac{q^i}{1-q}$$

We claim that $a_i = k \cdot S$.

$$k = \frac{a_i}{S} = q^{i-1} \cdot \frac{1-q}{q \cdot q^{i-1}} = \frac{1-q}{q}$$

This problem is solvable for $k \in (0, \infty) \because q < 1$.

21. Assume all x_i are nonzero real numbers, and that $(x_1^2 + x_2^2 + ... + x_{n-1}^2)(x_2^2 + x_3^2 + ... + x_n^2) = (x_1x_2 + x_2x_3 + ... + x_{n-1}x_n)^2$ holds for every $n \geq 3$. To prove $x_1, x_2, x_3, ..., x_n, ...$ forms an infinite geometric progression by mathematical induction, we need to first review the base case n=3.

$$LHS = (x_1^2 + x_2^2)(x_2^2 + x_3^2) = x_1^2x_2^2 + x_2^2x_3^2 + x_1^2x_3^2 + x_2^4$$

$$RHS = (x_1x_2 + x_2x_3)^2 = x_1^2x_2^2 + x_2^2x_3^2 + 2x_1x_2^2x_3$$

$$LHS = RHS \implies x_1^2x_3^2 - 2x_1x_2^2x_3 + x_2^4 = (x_1x_3 - x_2^2)^2 = 0$$

From the above result, we can conclude $x_1x_3 - x_2^2 = 0 \implies \frac{x_2}{x_1} = \frac{x_3}{x_2} = r$ where we define r as the common ratio of a GP. Hence the base case of n=3 gives us a geometric progression.

Now we assume that the statement $(x_1, x_2, x_3, ..., x_n, ...$ forms an infinite geometric progression) holds for all terms upto x_k where $(k \geq 3)$, that is $\frac{x_2}{x_1} = \frac{x_3}{x_2} = ... = \frac{x_k}{x_{k-1}} = r$. We need to prove it holds for $n = k+1$.

$$LHS = (x_1^2 + x_2^2 + ... + x_k^2)(x_2^2 + x_3^2 + ... + x_{k+1}^2)$$

$$RHS = (x_1x_2 + x_2x_3 + ... + x_kx_{k+1})^2$$

Let the first expression in parenthesis be written as $\sum a^2$ and the second in parenthesis be written as $\sum b^2$ in LHS and the expression in RHS as $(\sum ab)^2$. By Cauchy-Schwarz Inequality, $\left(\sum a^2\right)\left(\sum b^2\right) = \left(\sum ab\right)^2 \iff \frac{b_1}{a_1} = \frac{b_2}{a_2} = ...$ and substituting $a = x_i$ and $b = x_{i+1}$ we have $\frac{x_2}{x_1} = \frac{x_3}{x_2} = ... = \frac{x_{k+1}}{x_k}$. Since $\frac{x_2}{x_1} = r$ by the inductive hypothesis, this common value is r, so $\frac{x_2}{x_1} = \frac{x_3}{x_2} = ... = \frac{x_{k+1}}{x_k} = r$. Hence the case of $n = k+1$ gives us a geometric progression. By mathematical induction:

$$\frac{x_2}{x_1} = \frac{x_3}{x_2} = ... = r \implies (x_1^2 + x_2^2 + ... + x_{n-1}^2)(x_2^2 + x_3^2 + ... + x_n^2) = (x_1x_2 + x_2x_3 + ... + x_{n-1}x_n)^2$$

For the converse, if $x_{i+1} = rx_i$ for all i, then $b_i = ra_i$, so $\sum b^2 = r^2 \sum a^2$ and $\sum ab = r \sum a^2$, giving $\left(\sum a^2\right)\left(\sum b^2\right) = r^2\left(\sum a^2\right)^2 = \left(\sum ab\right)^2$. Hence the two statements are equivalent:

$$\frac{x_2}{x_1} = \frac{x_3}{x_2} = ... = r \iff (x_1^2 + x_2^2 + ... + x_{n-1}^2)(x_2^2 + x_3^2 + ... + x_n^2) = (x_1x_2 + x_2x_3 + ... + x_{n-1}x_n)^2$$